

The Story of Time - Sundials to the Atomic Second

UK Family Tour

The Story of Time: From Sundials to the Atomic Second
Everything you saw at Greenwich today was, in the end, about one question: what time is it, really? The red ball that drops from the Observatory turret at one o'clock, the brass meridian line, the room full of Harrison's clocks, the very phrase Greenwich Mean Time, all of it is the record of humanity's long struggle to measure time and, harder still, to make everyone agree on it. It is worth telling that story from the beginning, because it ends with the invisible machinery that runs your phone, your bank, and the satellites overhead tonight, and because so much of it was decided on this small hill.

For almost all of human history, time meant the sun. The first clock was a shadow. The Egyptians raised obelisks whose shadows swept across the ground through the day and built shadow clocks and sundials more than three thousand years ago. The logic was simple and universal: noon is the moment the sun stands highest and shadows are shortest, and the hours are slices of the sun's journey across the sky. The trouble with a sundial is obvious the moment a cloud passes or the sun sets, so people built clocks that did not need the sky. Water clocks, dripping steadily from one vessel to another, kept time in Egypt, Greece, China and the Islamic world for millennia; the Chinese engineer Su Song built a colossal water-driven astronomical clock tower in 1088. There were candle clocks marked in hours, and incense clocks that measured time by scent. But all of them drifted, and none could be trusted for long.

The great leap came in medieval Europe, and fittingly it came from the church. Monasteries needed to know when to ring the bells for prayer, and by the late thirteenth century mechanical clocks driven by falling weights, regulated by a clever oscillating mechanism called the verge-and-foliot escapement, were appearing in monastery and cathedral towers. These early clocks had no hands and no face; they simply struck a bell, and our word clock comes from the medieval word for bell. They were wildly inaccurate by our standards, losing or gaining perhaps fifteen minutes a day, but they were revolutionary, because for the first time time

was being kept by a machine indoors, independent of the sun, ticking on through the night.

For four centuries clocks slowly improved, but the next enormous jump came in 1656, when the Dutch scientist Christiaan Huygens built the first pendulum clock. Galileo had noticed that a swinging weight keeps remarkably steady time, and Huygens turned the insight into an instrument. Overnight, accuracy improved from about fifteen minutes a day to around fifteen seconds a day, a hundredfold gain. The tall longcase clocks, the grandfather clocks, that followed were the most accurate machines on Earth, and for the first time it made sense to put a minute hand on a clock at all. But a pendulum needs to hang still and level, which is why, as this morning's story of longitude made clear, the pendulum was useless at sea, and why John Harrison's spring-driven marine chronometers of the 1760s were such a triumph. By carrying accurate time in a box across the ocean, the chronometer did not just save ships; it proved that portable, mechanical timekeeping could be trusted to a fraction of a second a day.

And here we arrive at the strange truth that Greenwich forces you to confront. For all these centuries, even as clocks grew superb, there was no such thing as the time. Every town kept its own local solar time, its clocks set so that noon fell when the sun was highest over that particular spot. Noon in Bristol was about ten minutes later than noon in London, because Bristol is further west and the sun reaches it later. Nobody minded, because nobody could travel fast enough for it to matter. A coach took days; the sun set your watch when you arrived. It was the railway that broke this ancient arrangement. When trains could carry you from London to Bristol in a couple of hours, a timetable written in dozens of different local times became chaos, and a real danger, because two trains running on clocks ten minutes apart can collide. So the railways imposed order. The Great Western Railway adopted London time along its whole line in 1840, and other companies followed, and this new synchronized standard was actually called railway time. By the 1850s most of Britain had abandoned its local suns and set

its clocks to a single national standard, and that standard was the mean time kept at the Royal Observatory here at Greenwich. It became law across Great Britain in 1880.

Why Greenwich? Because Greenwich had spent two hundred years becoming the most trusted timekeeper in the world. The Observatory, founded in 1675, existed to measure the heavens precisely enough to solve navigation, and precise astronomy demands precise time. So Greenwich distributed its time outward. From 1833 the great red time ball you saw has dropped down its mast at exactly one o'clock every afternoon, a visual signal so that ships' captains on the Thames could set their chronometers to the instant before sailing. On the Observatory wall hangs the Shepherd Gate Clock, installed in 1852, which showed Greenwich time to the public on a twenty-four-hour dial and was one of the first clocks ever driven by electric pulses sent down a wire. Time was becoming a signal, something you could send by telegraph across a whole country in the blink of an eye, and Greenwich was the source.

The final step was to do for the world what the railways had done for Britain. If Britain needed one time, a planet crossed by telegraph cables and steamships needed a shared system too. At the International Meridian Conference in Washington in 1884, the same conference that made Greenwich the Prime Meridian of longitude, the nations agreed to build the world's time zones outward from Greenwich. The globe was divided into twenty-four zones, each fifteen degrees of longitude wide, each one hour apart, all counted from the meridian that runs through the Observatory courtyard. This is why, wherever you are on Earth tonight, your time zone is defined as so many hours ahead of or behind Greenwich. Greenwich Mean Time became the still point around which the whole turning world set its clocks.

But the story does not end with mechanical clocks and Victorian confidence, because the twentieth century discovered that the Earth itself is an unreliable clock. In 1927 the quartz clock arrived, using the steady electrical vibration of a quartz crystal, and it was so stable it could actually detect tiny irregularities in the Earth's rotation, wobbles and a gradual slowing that no one had been able to measure before. The ground beneath the whole system, the assumption that one day is a fixed length, turned out to be slightly false. So physicists went looking for a clock that did not depend on the Earth at all, and they found it in the atom. In 1955, at the National Physical Laboratory in England, Louis Essen and Jack Parry built the first accurate atomic clock, which

counts time by the fantastically regular vibration of caesium atoms, more than nine billion oscillations every second. It was so much better than the spinning Earth that in 1967 the world redefined the second itself: no longer a fraction of a day, but a fixed number of caesium vibrations, 9,192,631,770 of them. For the first time in history, time was cut loose from the sun and the stars.

This creates a delicate problem, and it is the reason the phrase Greenwich Mean Time is now, strictly speaking, slightly out of date. The atomic clocks keep perfect, unchanging time, but the Earth keeps wandering, spinning a hair slower over the centuries as the Moon's tides drag on it. So the world's official time, Coordinated Universal Time or UTC, is a compromise: it ticks with atomic precision, but every few years, when the gap between atomic time and the sun grows too wide, the world adds a leap second, holding the clocks back for one strange extra tick to let the sun catch up. Twenty-seven of these leap seconds have been quietly inserted since 1972. They are fiddly and they cause trouble for computers, and in 2022 the world's timekeepers voted to abolish them by 2035, finally letting atomic time and solar time drift gently apart. When people say GMT today, they almost always mean UTC, atomic time wearing Greenwich's old name.

And that atomic time is now the invisible heartbeat of modern life. The internet, the electricity grid, and the global financial system all depend on computers agreeing on the time to within thousandths of a second. Most of all, satellite navigation is simply timekeeping in disguise: each GPS satellite carries an atomic clock, and your phone finds its position by measuring, to within billionths of a second, how long signals take to arrive from several of them. It is so precise that engineers must correct for Einstein's relativity, because the clocks in orbit, moving fast and sitting higher in the Earth's weaker gravity, genuinely run about thirty-eight millionths of a second faster each day, and if that were ignored, GPS would drift you off course by miles within a day. The newest optical atomic clocks are so accurate they would not lose a single second over the entire age of the universe, and they may soon force us to redefine the second all over again.

So when you looked at the time ball dropping over Greenwich today, you were looking at the visible tip of an idea that runs from a shadow on Egyptian sand to a caesium atom humming in a laboratory. The sundial gave us the hours, the monks gave us the machine, Huygens gave us the pendulum, Harrison gave us portable time, the railways

gave us a shared time, Greenwich gave the world its zero, and the atom gave us a second so perfect it no longer needs the sky at all. The whole of that history is why, in a small

park in south-east London, you can stand on the line where the world agreed to start counting time.